Supporting Information

ChemIP: An Open-Source Web Platform for Unified Chemical Safety Data Retrieval across Korean and International Regulatory Sources

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S1. Platform Installation and Configuration

Prerequisites

- Python 3.11+ with pip
- Node.js 18+ with npm
- (Optional) Ollama for local LLM features

Step 1: Clone and install dependencies

```
git clone https://github.com/yuyongkim/chemIP.git
cd chemIP

# Backend
pip install -r requirements.txt

# Frontend
cd frontend && npm install && cd ..
```

Step 2: Configure environment variables

Copy `.env.example` to `.env` and provide the following API keys:

Variable	Source	Required
`KOSHA_SERVICE_KEY_DECODED`	Korea Occupational Safety and Health Agency	Yes
`KIPRIS_API_KEY`	Korea Intellectual Property Rights Information Service	Yes
`KOTRA_API_KEY_DECODED`	Korea Trade-Investment Promotion Agency	Yes
`DRUG_API_KEY_DECODED`	Ministry of Food and Drug Safety (MFDS)	Yes
`PUBMED_API_KEY`	NCBI E-utilities	No
`NAVER_CLIENT_ID` / `NAVER_CLIENT_SECRET`	Naver Developers (news fallback)	No

All Korean government API keys are available at no cost through the respective open data portals (data.go.kr for KOSHA/MFDS, kipris.or.kr for KIPRIS, openapi.kotra.or.kr for KOTRA).

Step 3: Start services

```
# Backend (FastAPI)
uvicorn backend.main:app --host 127.0.0.1 --port 7010

# Frontend (Next.js)
cd frontend && npm run dev
```

Access points after startup:

Service	URL
Web interface	http://localhost:7000
API server	http://localhost:7010
Interactive API docs (Swagger)	http://localhost:7010/docs
Health check	http://localhost:7010/health

Step 4: Production deployment (optional)

For persistent operation, a PM2 process manager configuration is provided:

```
# Windows
start_all.bat

# Linux/macOS
bash start_all.sh
```

S2. API Endpoint Reference

ChemIP exposes a RESTful API organized into seven route modules. All endpoints are prefixed with `/api/`.

S2.1 Chemical Substance Endpoints (`/api/chemicals`)

Method	Endpoint	Description
GET	`/`	Full-text search by chemical name or CAS number (paginated)
GET	`/autocomplete`	Typeahead suggestions for chemical names
GET	`/{chem_id}`	Retrieve complete MSDS data (all 16 sections) with GHS classification
GET	`/{chem_id}/drugs`	Linked pharmaceutical data for a chemical (MFDS/OpenFDA/PubMed)

When a chemical detail is requested and MSDS sections are not cached locally, the system fetches all 16 sections from the KOSHA API in parallel and stores them for subsequent requests.

S2.2 Patent Endpoints (`/api/patents`)

Method	Endpoint	Description
GET	`/`	Search Korean patents via KIPRIS keyword query
GET	`/kipris/{application_number}`	Retrieve detailed KIPRIS patent record
GET	`/uspto/{chem_id}`	Search local USPTO patent index by chemical identifier
GET	`/global/{chem_id}`	Search global patent index (USPTO/EPO/WIPO/CN/JP/KR)

The global patent index contains 350M+ patent records stored in a local SQLite database (134.5 GB), enabling sub-second full-text search without external API calls.

S2.3 Drug and Pharmaceutical Endpoints (`/api/drugs`)

Method	Endpoint	Description
GET	`/approval`	MFDS drug approval database search
GET	`/easy`	MFDS consumer medication information
GET	`/openfda`	US drug labels by brand, generic, or substance name
GET	`/pubmed`	Biomedical literature search (multi-query: base + toxicology + safety)
GET	`/unified`	Parallel search across all three sources (MFDS + OpenFDA + PubMed)

S2.4 Trade Intelligence Endpoints (`/api/trade`)

Method	Endpoint	Description
GET	`/news`	KOTRA overseas market news
GET	`/strategy`	Market entry strategy reports by country
GET	`/prices`	Product price trend information
GET	`/fraud`	Trade fraud case database
GET	`/import-regulations`	Import restrictions by country
GET	`/naver-news`	Korean news search (Naver, fallback source)

Additional endpoints exist for country information, enterprise success cases, and tourism data.

S2.5 Safety Guide Endpoints (`/api/guides`)

Method	Endpoint	Description
GET	`/status`	KOSHA guide dataset availability and statistics
GET	`/search`	Full-text search across KOSHA safety guides
GET	`/{guide_no}`	Retrieve detailed KOSHA safety guide by ID
GET	`/recommend/{chem_id}`	Recommended safety guides for a chemical (multi-criteria matching)

S2.6 AI Analysis Endpoints (`/api/ai`)

These endpoints require a local Ollama LLM server (default model: qwen3:8b).

Method	Endpoint	Description
POST	`/analyze`	Multi-source safety analysis report (MSDS + patents + guides)
POST	`/summarize`	Comprehensive safety summary across all data sources
POST	`/drug-analysis`	Drug-chemical relationship analysis
POST	`/recommend`	Related chemical/keyword recommendations
POST	`/ask`	Natural language Q&A with full data context
GET	`/llm-status`	Check LLM server availability

S2.7 Regulatory Classification Endpoints (`/api/regulations`)

Source	Endpoints	Data Provided
ECHA (EU)	`/echa/search`, `/echa/clp/{id}`, `/echa/labelling/{id}`	REACH registration, CLP classification, harmonised labelling
EPA CompTox (US)	`/comptox/search`, `/comptox/hazard/{dtxsid}`	ToxValDB hazard data, cancer classification, functional use
NIOSH (US)	`/niosh/search`, `/niosh/cas/{cas}`, `/niosh/exposure/{cas}`	REL/PEL/IDLH exposure limits, PPE recommendations, carcinogen list
KISCHEM (KR)	`/kischem/search`, `/kischem/cas/{cas}`	Exposure symptoms, first-aid procedures
NCIS (KR)	`/ncis/cas/{cas}`, `/ncis/search`	Korean environmental substance classification
Aggregated	`/search`	Cross-source search (all five sources in parallel)

Total: 80+ API endpoints across all modules.

S3. Database Schema

ChemIP uses three local SQLite databases:

S3.1 terminology.db (52.6 MB) — Primary chemical and MSDS database

Table: `chemical_terms` (117,744 rows)

Column	Type	Description
id	INTEGER PK	Internal row ID
name	TEXT	Chemical name (Korean or mixed)
cas_no	TEXT	CAS registry number
name_en	TEXT	English name (when available)
source	TEXT	Data source: KOSHA, ECHA, KREACH, KISCHEM, PUBCHEM
external_id	TEXT	External source identifier

Source distribution: ECHA 80,534 (68.4%), KREACH 21,804 (18.5%), KOSHA 13,448 (11.4%), KISCHEM 1,954 (1.7%), PubChem 4 (<0.1%).

Table: `msds_details` (550,161 rows)

Column	Type	Description
chem_id	TEXT	Chemical identifier
section_no	INTEGER	MSDS section number (1-16)
xml_data	TEXT	XML-formatted section content from KOSHA API

Table: `msds_english` (56,176 rows)

Column	Type	Description
chem_id	TEXT	Chemical identifier
cas_no	TEXT	CAS registry number
name_en	TEXT	English chemical name
pubchem_cid	INTEGER	PubChem compound ID
signal_word	TEXT	GHS signal word (Danger/Warning)
ghs_classification	TEXT	JSON array of GHS classifications
hazard_statements	TEXT	JSON array of H-statements
precautionary_statements	TEXT	JSON array of P-statements
pictograms	TEXT	JSON array of GHS pictogram codes

Table: `chemical_terms_fts` — FTS5 virtual table for full-text search over name, cas_no, name_en.

S3.2 global_patent_index.db (134.5 GB) — Global patent full-text index

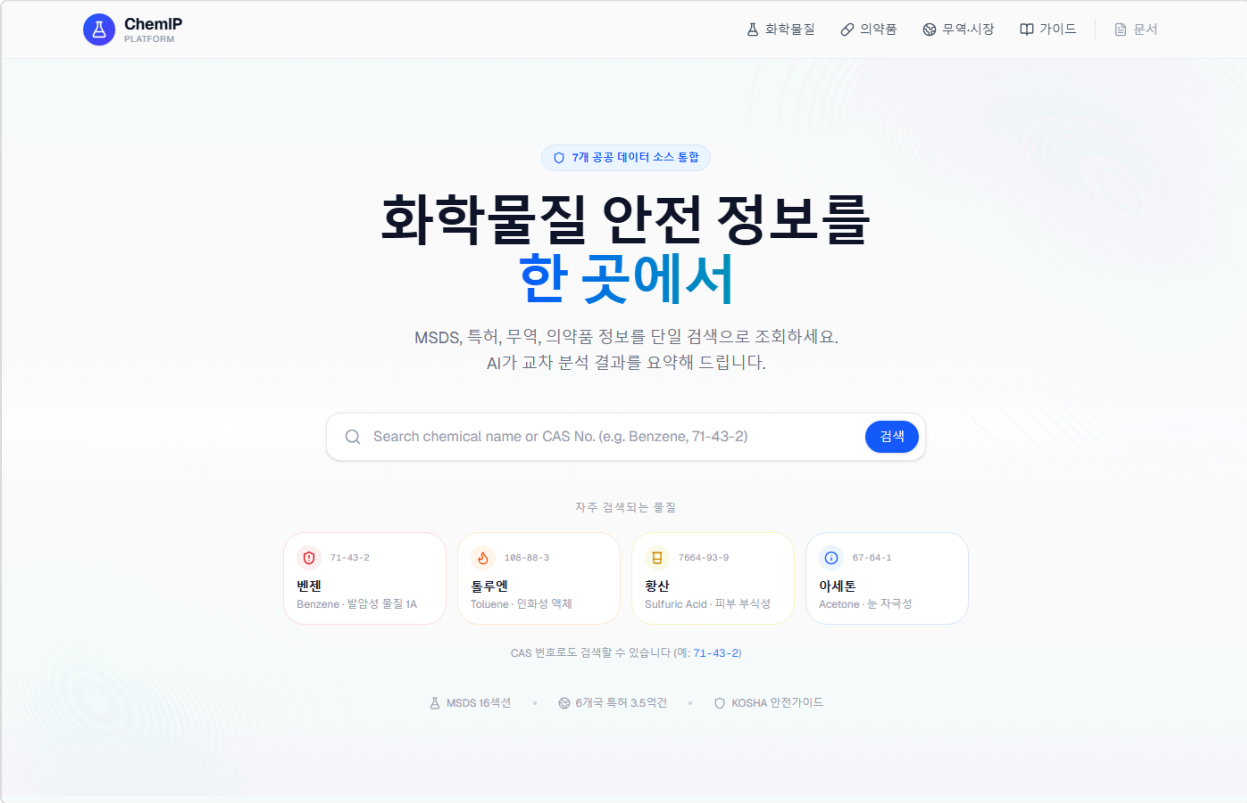
Contains patent records from USPTO, EPO, WIPO, CN, JP, and KR patent offices with chemical name matching via FTS5 indexing.

S3.3 uspto_index.db (5.8 GB) — US patent index

Dedicated USPTO patent index for faster US-specific patent queries.

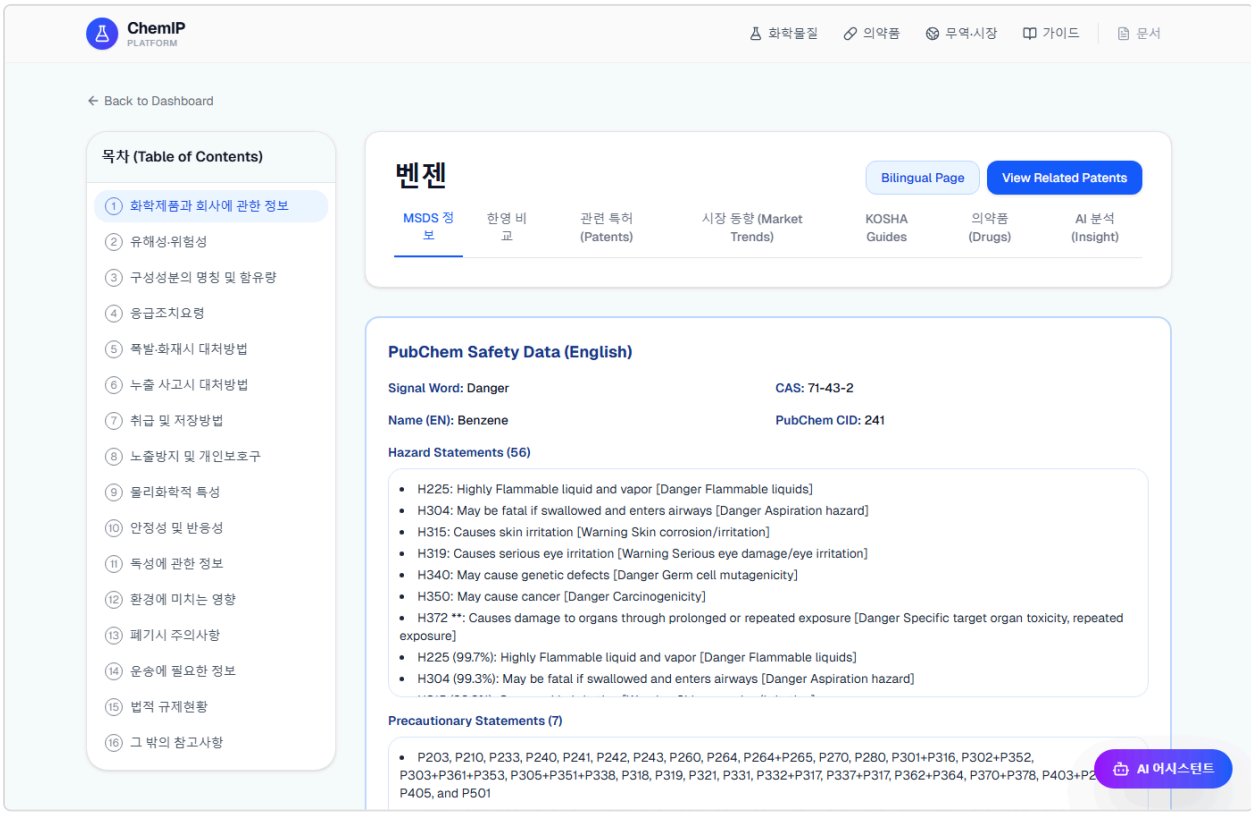
S4. User Interface Screenshots

Figure S1. Home page with integrated chemical and drug search



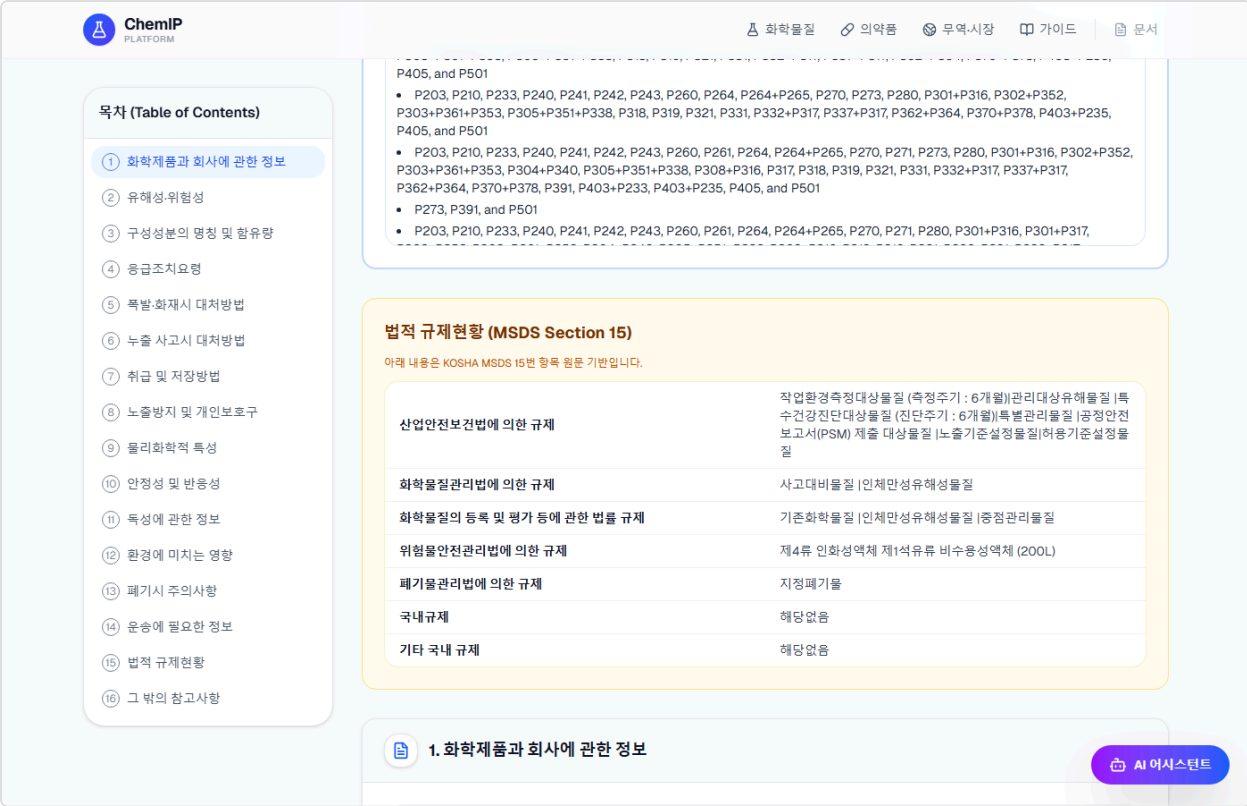
The home page presents a unified search interface with KOSHA MSDS as the primary entry point. Statistics display 13,448 indexed KOSHA chemicals with full 16-section MSDS coverage. Quick-access cards provide popular chemical substances with GHS hazard categories.

Figure S2. Chemical detail page with MSDS sections



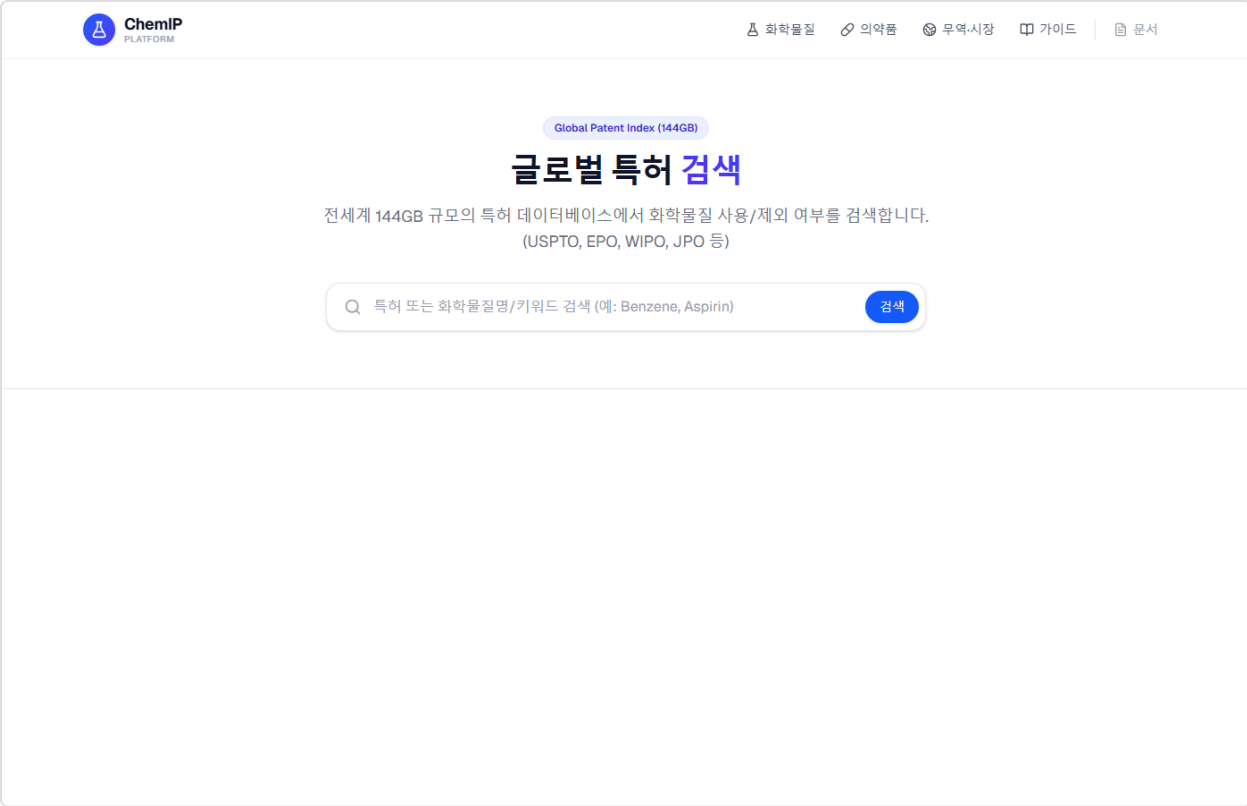
The chemical detail page displays the full 16-section MSDS data retrieved from KOSHA, organized with a table-of-contents sidebar for navigation. The English safety summary (sourced from PubChem GHS data) appears at the top, followed by Korean MSDS sections including GHS pictograms, hazard statements, and handling/storage procedures.

Figure S3. Chemical detail page — bilingual safety view and regulation status



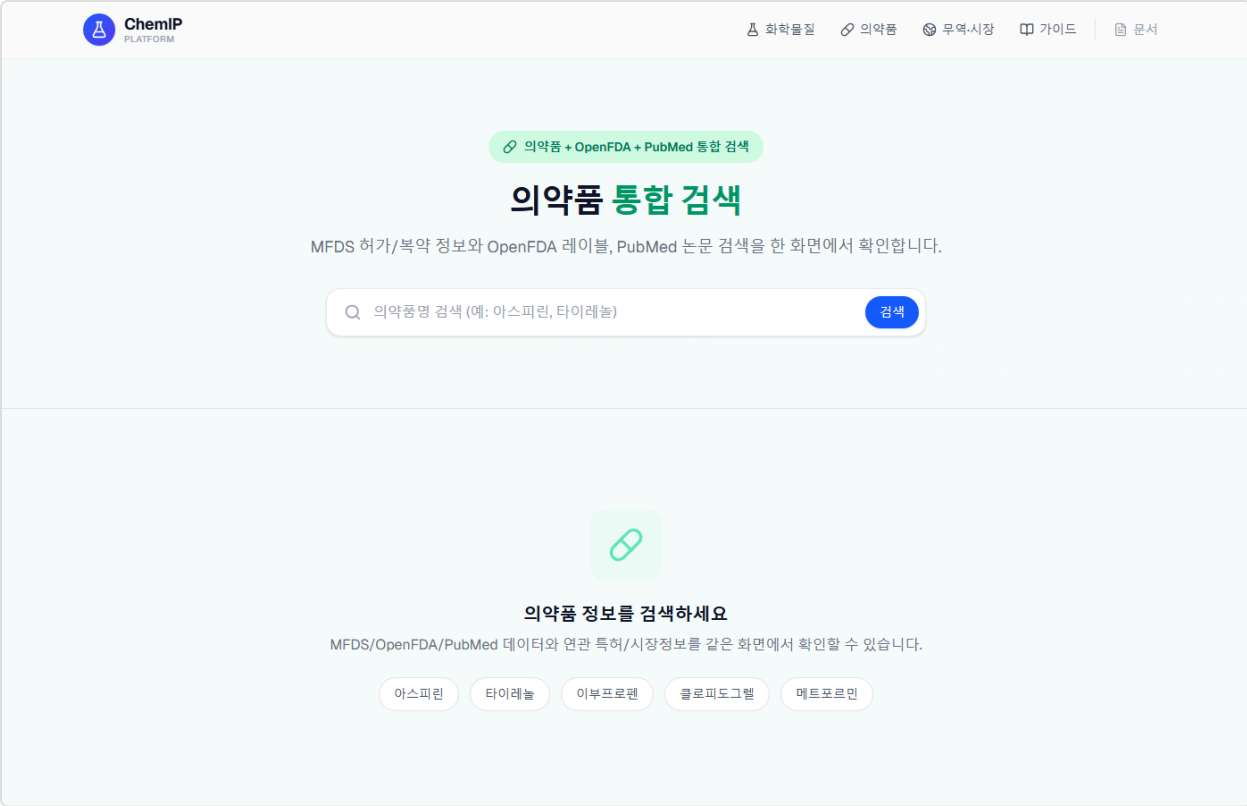
An alternative view of the chemical detail page showing the bilingual safety panel (Korean KOSHA + English PubChem side-by-side) and Korean regulatory classification status from NCIS and KISCHEM.

Figure S4. Patent search results with chemical highlighting



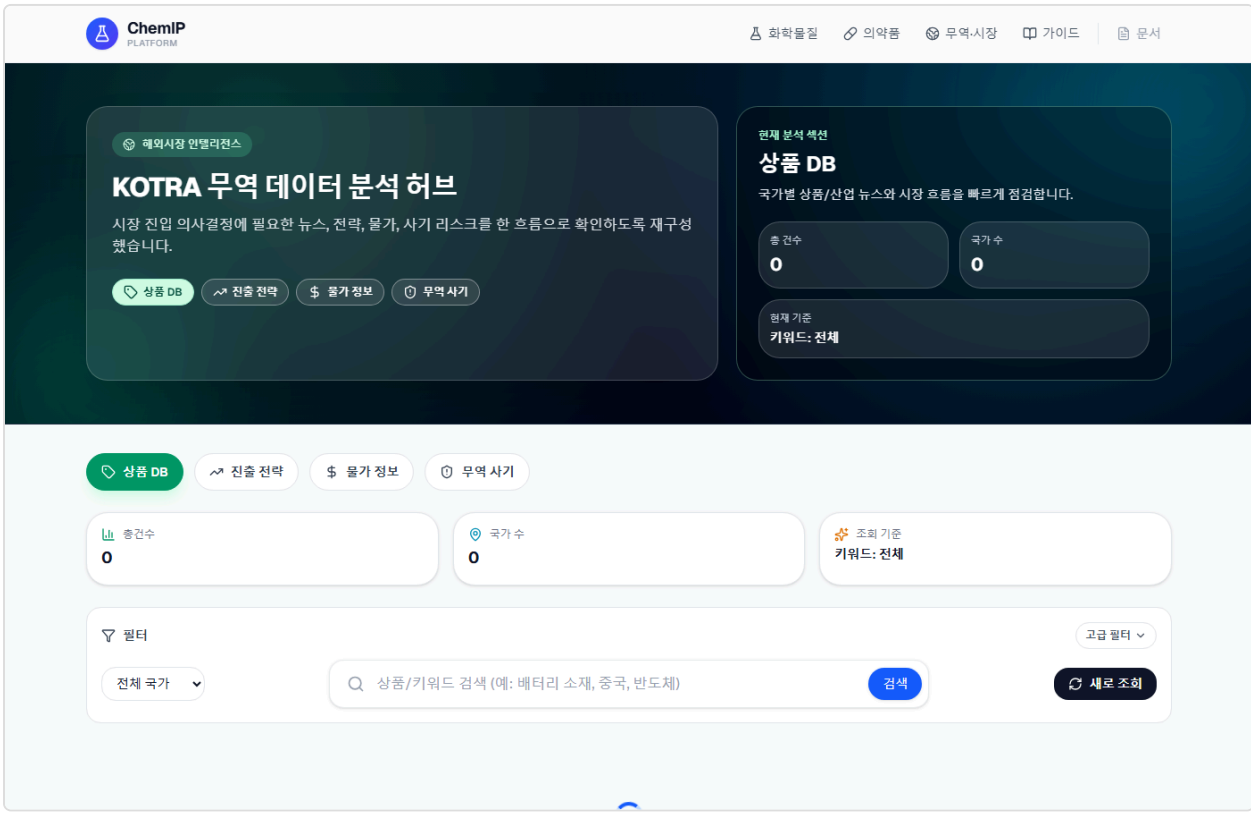
The patent viewer displays results from both the local global patent index (350M+ records from 6 jurisdictions) and KIPRIS real-time search. Chemical names and CAS numbers are highlighted in patent titles and abstracts.

Figure S5. Drug cross-reference panel



The unified drug search returns results from three sources in parallel: MFDS (Korean drug approvals and consumer medication information), OpenFDA (US drug labels), and PubMed (biomedical literature). Results are grouped by source with direct links to original records.

Figure S6. Trade intelligence dashboard



The trade data hub integrates KOTRA market news, entry strategies, price trends, and trade fraud cases. Filters support country, keyword, and date range selection with real-time result counts.

Figure S7. KOSHA safety guide recommendations

운영 가이드 (Operations Guide)

실제 사용 및 배포를 위한 가이드입니다.

Git에 올려야 하는 것 / 올리면 안 되는 것

✅ Git에 올려야 하는 것

```
MSDS/
├── backend/  # 백엔드 소스 코드
├── frontend/  # 프론트엔드 소스 코드
├── scripts/  # 인덱싱 스크립트
├── tests/  # 테스트 코드
├── docs/  # 문서
├── .gitignore  # Git 제외 설정
├── README.md  # 프로젝트 설명
└── requirements.txt  # Python 의존성
```

❌ Git에 올리면 안 되는 것

```
data/  # 140GB DB 파일들
.venv/  # Python 가상환경
node_modules/  # npm 패키지
.env  # API 키 등 비밀 정보
*.db  # 데이터베이스 파일
*.log  # 로그 파일
```

140GB 데이터베이스 공유 방법

방법	장점	단점
1. NAS/공유 드라이브	팀 내부에서 직접 접근 가능	외부 접근 불가
2. 클라우드 (S3, GCS)	어디서든 다운로드 가능	비용 발생
3. 외장하드 전달	가장 빠른 전송	물리적 전달 필요
4. 로컬에서 재생성	항상 최신 데이터	24-48시간 소요

The safety guides tab recommends relevant KOSHA occupational safety guides for the queried chemical using multi-criteria matching (chemical name, CAS number, hazard category, and industry sector).

S5. Deployment Modes

ChemIP supports three deployment configurations to accommodate different resource constraints:

Mode	Components	Database	Storage	Use Case
Minimal	Backend + Frontend	terminology.db only (53 MB)	~200 MB	MSDS lookup and drug search
Standard	+ KOSHA Guides + Ollama LLM	+ guide dataset (~500 MB)	~4 GB	Full safety analysis with AI
Full	+ Global patent index	+ patent DBs (140 GB)	~145 GB	Complete platform with patent search

All modes support the same API interface; unavailable features return graceful error messages rather than failing silently, consistent with the adapter-based graceful degradation design described in the main text.

This Supporting Information accompanies the manuscript "ChemIP: An Open-Source Web Platform for Unified Chemical Safety Data Retrieval across Korean and International Regulatory Sources" submitted to ACS Chemical Health & Safety.